1. Hospital Patient Billing System

Develop a Java program for a hospital billing system using Class and Objects.

Code:

```
import java.util.Scanner;
class Hospital {
    int pid, days;
    String pname, room;
    double treatment;
    void read() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Patient ID: ");
        pid = sc.nextInt();
        sc.nextLine();
        System.out.print("Patient Name: ");
        pname = sc.nextLine();
        System.out.print("Room Type (General/Special): ");
        room = sc.nextLine();
        System.out.print("Days Admitted: ");
        days = sc.nextInt();
        System.out.print("Treatment Charges: ");
        treatment = sc.nextDouble();
    }
    void bill() {
        double roomCharge;
        if(room.equalsIgnoreCase("General"))
            roomCharge = days * 1500;
        else
            roomCharge = days * 3000;
```

```
        double total = roomCharge + treatment;
        System.out.println("Total Bill = " + total);
    }
    public static void main(String args[]) {
        Hospital h = new Hospital();
        h.read();
        h.bill();
    }
}
```

Output:

```
Patient ID: 101
Patient Name: Ravi
Room Type (General/Special): General
Days Admitted: 4
Treatment Charges: 5000
Total Bill = 11000.0
```

2. Airline Ticket Reservation System

Develop a Java program for an airline reservation system using Class and Objects.

Code:

```
import java.util.Scanner;
class Airline {
    int pid, tickets;
    String pname, flight;
    double fare;
    void read() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Passenger ID: ");
        pid = sc.nextInt();
```

```
        sc.nextLine();
        System.out.print("Passenger Name: ");
        pname = sc.nextLine();
        System.out.print("Flight Number: ");
        flight = sc.nextLine();
        System.out.print("Ticket Fare: ");
        fare = sc.nextDouble();
        System.out.print("Number of Tickets: ");
        tickets = sc.nextInt();
    }
    void calculate() {
        double total = fare * tickets;
        if(tickets > 5)
            total = total - (total * 0.05);
        total = total + (total * 0.12);
        System.out.println("Final Amount = " + total);
    }
    public static void main(String args[]) {
        Airline a = new Airline();
        a.read();
        a.calculate();
    }
}
```

Output:

```
Passenger ID: 1
Passenger Name: Sai
Flight Number: AI101
Ticket Fare: 5000
Number of Tickets: 6
Final Amount = 31920.0
```

3. Vehicle Rental Management System

Develop a Java program for a vehicle rental company using Class and Objects.

Code:

```
import java.util.Scanner;
class VehicleRental {
    int cid, days;
    String cname, type;
    void read() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Customer ID: ");
        cid = sc.nextInt();
        sc.nextLine();
        System.out.print("Customer Name: ");
        cname = sc.nextLine();
        System.out.print("Vehicle Type (Car/Bike): ");
        type = sc.nextLine();
        System.out.print("Rental Days: ");
        days = sc.nextInt();
    }
    void bill() {
        double amount;
        if(type.equalsIgnoreCase("Car"))
            amount = days * 2000;
        else
            amount = days * 500;
        amount += 500; // penalty
        System.out.println("Final Bill = " + amount);
    }
```

```
    public static void main(String args[]) {
        VehicleRental v = new VehicleRental();
        v.read();
        v.bill();
    }
}
```

Output:

Customer ID: 1001

Customer Name: Kumar

Vehicle Type (Car/Bike): Car

Rental Days: 3

Final Bill = 6500.0

4. Manufacturing Production Monitoring System

Develop a Java program to monitor daily production in a manufacturing company using Class and Objects.

Code:

```
import java.util.Scanner;
class Production {
    int pid, target, produced;
    String pname;
    void read() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Product ID: ");
        pid = sc.nextInt();
        sc.nextLine();
        System.out.print("Product Name: ");
        pname = sc.nextLine();
        System.out.print("Target Quantity: ");
        target = sc.nextInt();
```

```
        System.out.print("Produced Quantity: ");
        produced = sc.nextInt();
    }
    void status() {
        double efficiency = (double)produced / target * 100;
        System.out.println("Efficiency = " + efficiency + "%");
        if(produced >= target)
            System.out.println("Status: Achieved");
        else
            System.out.println("Status: Not Achieved");
    }
    public static void main(String args[]) {
        Production p = new Production();
        p.read();
        p.status();
    }
}
```

Output:

Product ID: 101

Product Name: Mobile

Target Quantity: 1000

Produced Quantity: 1100

Efficiency = 110.0%

Status: Achieved

5. Online Course Enrollment System

Develop a Java program for an online learning platform using Class and Objects.

Code:

```
import java.util.Scanner;

class Course {

    int sid;

    String sname, cname;

    double fee;

    void read() {

        Scanner sc = new Scanner(System.in);

        System.out.print("Student ID: ");

        sid = sc.nextInt();

        sc.nextLine();

        System.out.print("Student Name: ");

        sname = sc.nextLine();

        System.out.print("Course Name: ");

        cname = sc.nextLine();

        System.out.print("Course Fee: ");

        fee = sc.nextDouble();

    }

    void payment() {

        double amount = fee + 500;

        if(fee > 20000)

            amount = amount - (fee * 0.15);

        System.out.println("Payable Amount = " + amount);

    }

    public static void main(String args[]) {

        Course c = new Course();

        c.read();
```

```
        c.payment();
    }
}
```

Output:

Student ID: 101

Student Name: Mounika

Course Name: Java

Course Fee: 25000

Payable Amount = 21750.0